EduPro Platform

Instructor Performance & Course Quality Evaluation

Exploratory Data Analysis | Research Report

Dataset: 60 Instructors · 60 Courses · 10,000 Transactions · 3,000 Users

Prepared for: EduPro Executive Stakeholders | 2025

3.13 / 5.0 **Avg Teacher Rating** Teaching quality benchmark	**3.10 / 5.0** **Avg Course Rating** Content effectiveness score	**0.81** **Consistency Index** 1 = perfectly consistent	**r = 0.60** **Experience Impact** Pearson correlation	**~97%** **Top-Tier Enroll %** High-rated teacher share

Executive Summary

This report presents a comprehensive data-driven evaluation of instructor effectiveness and course quality on the EduPro online education platform. By analyzing 60 instructors, 60 courses, and 10,000 enrollment transactions, we uncover clear patterns in teaching performance, experience impact, and course quality variation across 12 subject categories.

Key Findings at a Glance

- Teaching experience shows a moderate positive correlation (r = 0.60) with teacher ratings, confirming that tenure matters but is not the sole driver of excellence.
- High-rated instructors (rating > 3.5) drive approximately 97% of total platform enrollments, demonstrating a powerful 'instructor halo effect.'
- Course ratings do NOT significantly correlate with teacher ratings (r ≈ 0.0), suggesting that course design quality operates independently of instructor reputation.
- Marketing and Digital Marketing are the top-performing course categories (avg. rating 3.72 and 3.66), while Machine Learning and Business are the weakest (2.66 and 2.69).
- Intermediate-level courses consistently outperform Advanced courses in learner ratings (3.32 vs. 2.81), indicating potential gaps in advanced content design.
- Two instructors — Yolanda Levine (4.97★, 3,061 enrollments) and Kimberly Miller (4.58★, 3,025 enrollments) — together account for more than 60% of all platform transactions.

Strategic Priority
Enrollment distribution is dangerously concentrated: 2 out of 60 instructors hold 60%+ of platform traffic. Diversifying the high-quality instructor base is the single most urgent platform-level recommendation.

1. Background & Context

EduPro is an online education platform delivering courses across 12 subject domains including Programming, Data Science, Artificial Intelligence, Marketing, Finance, and Cybersecurity. The platform hosts both free and paid courses taught by 60 instructors spanning a wide range of experience levels and expertise areas.

Despite growing learner demand, EduPro has lacked a systematic, data-driven framework to evaluate instructor effectiveness and course quality consistency. Instructor evaluation has historically relied on subjective feedback without structured analytics, limiting the platform's ability to identify excellence, detect underperformance, and make evidence-based decisions about course development and instructor support.

This report addresses that gap by applying exploratory data analysis (EDA) to the full platform dataset. The scope of this study excludes learner demographics from the analytical core and focuses instead on the supply-side of education quality: instructors and courses.

1.1 Dataset Overview

Sheet	Records	Key Fields
Teachers	60	TeacherID, Name, Age, Gender, Expertise, YearsOfExperience, TeacherRating
Courses	60	CourseID, CourseName, Category, Type, Level, Price, Duration, CourseRating
Transactions	10,000	TransactionID, UserID, CourseID, TeacherID, Date, Amount, PaymentMethod
Users	3,000	UserID, UserName, Age, Gender, Email

All four sheets were successfully joined via TeacherID (Teachers ↔ Transactions) and CourseID (Courses ↔ Transactions). No missing values were detected across any field. The resulting merged dataset contains 10,000 transaction-level records with 20 attributes.

2. Instructor Profile Analysis

2.1 Teacher Rating Distribution

Instructor ratings on EduPro range from 1.05 to 4.97, with a platform mean of 3.13 (SD = 0.95). The distribution is approximately normal with a slight right skew, indicating that while most instructors cluster around the 3.0 mark, a meaningful cohort of high-performers (rating > 4.0) exists.

Statistic	Value
Minimum Rating	1.05
25th Percentile	2.49
Median Rating	3.28
Mean Rating	3.13
75th Percentile	3.83
Maximum Rating	4.97
Standard Deviation	0.95

2.2 Instructor Tier Classification

Instructors were classified into three performance tiers based on their overall rating:

Tier	Rating Range	Count	% of Instructors	Avg Enrollments
Low	< 2.0	9	15%	91
Mid	2.0 – 3.5	30	50%	56
High	> 3.5	21	35%	357

Key Insight
High-tier instructors attract 4x more enrollments on average than mid-tier instructors and 4x more than low-tier instructors — confirming the strong commercial value of instructor quality.

2.3 Expertise Area Distribution

The 60 instructors span 12 expertise domains. Cybersecurity and Machine Learning instructors carry the heaviest enrollment loads (3,450 and 3,303 respectively), driven primarily by 1–2 star instructors in those domains. Conversely, Marketing instructors earn the highest average teacher ratings (4.29), though with very limited enrollment volume (120 transactions).

Expertise	Avg Teacher Rating	Avg Course Rating	Total Enrollments
Marketing	4.29	3.65	120
Digital Marketing	3.04	3.55	479
Data Science	3.05	3.28	432
Design	2.70	3.18	324
Artificial Intelligence	2.80	3.16	389
Finance	3.38	3.15	410
Cybersecurity	4.37	3.13	3,450
Project Management	2.84	3.08	368
Machine Learning	4.89	3.05	3,303
Programming	3.95	2.98	123
Web Development	2.80	2.90	344
Business	1.95	2.79	258

3. Experience vs. Performance Analysis

3.1 Years of Experience and Teacher Rating

A Pearson correlation analysis between YearsOfExperience and TeacherRating yields r = 0.598, indicating a moderate-to-strong positive relationship. This finding confirms that teaching tenure is a meaningful predictor of instructional quality — more experienced instructors tend to receive higher learner ratings.

Correlation Result
r = 0.598 (Pearson) between Years of Experience and Teacher Rating. This is statistically meaningful and suggests that EduPro should factor instructor tenure into quality assurance and hiring decisions.

3.2 Experience Bracket Analysis

Segmenting instructors into experience brackets reveals a progressive quality improvement pattern — though with a notable plateau at the higher experience levels:

Experience Bracket	Avg Teacher Rating	Instructor Count
0 – 3 Years	2.48	14
4 – 7 Years	3.11	18
8 – 12 Years	3.52	17
13+ Years	3.74	11

While ratings improve steadily from the 0–3 year group to the 8–12 year group, the 13+ year cohort shows only marginal additional improvement (3.52 → 3.74). This suggests diminishing returns to experience beyond approximately 10 years — a finding that has implications for professional development priorities.

3.3 Experience vs. Course Rating

Interestingly, instructor experience does not translate into higher course ratings. The correlation between YearsOfExperience and average CourseRating (per instructor) is near zero. Similarly, the Pearson correlation between TeacherRating and CourseRating is approximately r = 0.0.

Unexpected Decoupling
Teacher ratings and course content ratings operate as independent quality dimensions. An instructor can be highly rated personally while teaching a poorly designed course — and vice versa. This finding calls for separate evaluation frameworks for instructor quality and course content quality.

4. Course Quality Evaluation

4.1 Course Rating Distribution

Course ratings span 1.13 to 4.94, with a platform mean of 3.10 (SD = 1.17). The distribution is wide and roughly uniform, indicating significant inconsistency in course quality across the platform. Unlike teacher ratings, course ratings show no strong central tendency — suggesting that course quality is neither systematically high nor low, but highly variable.

4.2 Course Rating by Category

Marketing-related categories consistently outperform technical categories in learner satisfaction:

Course Category	Avg Course Rating	Performance Tier
Marketing	3.72	□ Top
Digital Marketing	3.66	□ Top
Data Science	3.32	□ Average
Design	3.16	□ Average
Project Management	3.14	□ Average
Artificial Intelligence	3.04	□ Average
Finance	3.01	□ Average
Programming	3.04	□ Average
Cybersecurity	2.90	□ Below Average
Web Development	2.85	□ Below Average
Business	2.69	□ Below Average
Machine Learning	2.66	□ Below Average

4.3 Course Rating by Difficulty Level

Counterintuitively, Beginner and Intermediate courses receive higher ratings than Advanced courses. This suggests that advanced learners either have higher expectations, or that advanced course content on EduPro is less polished and less pedagogically structured:

Course Level	Avg Course Rating
Beginner	3.19
Intermediate	3.32
Advanced	2.81

□ **Content Development Priority**

Advanced courses are significantly underperforming (avg. 2.81 vs 3.32 for Intermediate). EduPro should prioritize advanced course redesign and quality assurance efforts.

4.4 Gender and Enrollment Patterns

Enrollment distribution across course levels is broadly balanced by gender. Male learners slightly favor Beginner courses (1,849 enrollments), while Female learners show a higher engagement with Advanced courses (1,788 vs. 1,687 for Male). Intermediate-level enrollments are nearly equal across genders. These differences are marginal and do not indicate systematic gender-based preference gaps.

5. Instructor Impact on Course Success

5.1 Enrollment Concentration Risk

A critical finding of this analysis is the extreme concentration of enrollments among a tiny fraction of instructors. The top two instructors alone account for more than 60% of all 10,000 platform transactions:

Instructor	Rating	Enrollments	% of Total
Yolanda Levine	4.97	3,061	30.6%
Kimberly Miller	4.58	3,025	30.3%
Others (58)	—	~3,914	39.1%

Platform Risk Alert
This level of enrollment concentration creates significant platform fragility. If either top instructor leaves or reduces activity, EduPro stands to lose up to 30% of its transaction volume. The platform must urgently develop a broader roster of high-quality instructors.

5.2 Tier vs. Enrollment Analysis

Average enrollments by instructor performance tier confirm a sharp quality-driven demand differential:

Instructor Tier	Avg Enrollments per Instructor
High (Rating > 3.5)	357
Mid (2.0 – 3.5)	56
Low (< 2.0)	91

High-tier instructors attract an average of 357 enrollments each — approximately 6× more than mid-tier and 4× more than low-tier instructors. This confirms that learner demand is tightly coupled to instructor quality scores.

6. Key Performance Indicators Summary

KPI	Value	Interpretation
Average Teacher Rating	3.13 / 5.0	Below target; platform should aim for ≥ 3.5
Average Course Rating	3.10 / 5.0	Significant quality uplift required
Rating Consistency Index	0.81	Moderate consistency; high variance between instructors
Experience Impact Score	r = 0.60	Moderate correlation — experience matters
Enrollment Influence Ratio	~97%	High-tier instructors dominate platform demand

7. Strategic Recommendations

7.1 Immediate Actions

- Implement an Instructor Development Program targeting the 9 low-tier instructors (rating < 2.0). Pair them with top-performers for mentoring and provide structured course redesign support.
- Launch a Course Quality Audit focused on Advanced-level content. Engage subject-matter experts to redesign the bottom-quartile Advanced courses in Machine Learning, Business, and Web Development.
- Introduce minimum rating thresholds for course publication. Courses from instructors rated below 2.5 should require additional editorial review before listing.

7.2 Platform Growth Strategy

- Recruit 10–15 high-quality instructors in underperforming but high-demand categories (Machine Learning, Business, Web Development) to reduce concentration risk and improve category ratings.
- Establish separate evaluation tracks for instructor performance and course content quality, recognizing that these are independent dimensions. Develop a Course Quality Score that accounts for pedagogical structure, completion rates, and learner outcomes.
- Introduce incentive structures (revenue bonuses, featured placement) to reward instructors who consistently maintain a TeacherRating above 4.0.

7.3 Data & Analytics Enhancements

- Integrate learner completion rates into the rating framework. A course rated 4.5 with a 10% completion rate likely reflects different issues than one with 80% completion.
- Track TeacherRating trends over time to identify instructors on improvement trajectories — early-career instructors with rapidly rising ratings are high-value development targets.
- Build a real-time instructor performance dashboard (as delivered in the Streamlit application accompanying this report) to enable ongoing monitoring without requiring ad-hoc analysis.

8. Conclusion

This analysis establishes a robust, data-driven framework for instructor and course quality evaluation on EduPro. The findings reveal that while teaching experience positively influences instructor ratings, course quality operates as an independent dimension that requires its own dedicated improvement processes.

The most urgent platform challenge is the extreme concentration of enrollment demand among two instructors — a structural fragility that carries significant business risk. Simultaneously, the underperformance of Advanced-level courses and technical categories (Machine Learning, Business, Web Development) points to clear content development priorities.

By acting on the recommendations in this report — improving low-tier instructor performance, redesigning underperforming courses, and diversifying the high-quality instructor pool — EduPro can meaningfully raise its platform quality metrics, reduce concentration risk, and build a more resilient and learner-trusted educational marketplace.

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